GitHub-Discord Bot

An AI-powered system to boost participation, motivation, and collaboration in remote teams

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Date: December 2nd, 2025
INFO 1170/5170: AI in Organizations

Abstract / Executive Summary

Our bot gamifies GiHub collaborations that motivate even lurkers to participate in the team work. It connects GitHub repository an Discord channels, and then rewards asynchronous contributions with points and badges in the team channel. Leveraging Gemini 2.5 flash model to generate personalized challenges and do sentiment analysis of messages for containing health communication.

- **AI-powered Discord bot**
- **Gamified contributions with rewards**
- **Stronger engagement in remote teams**

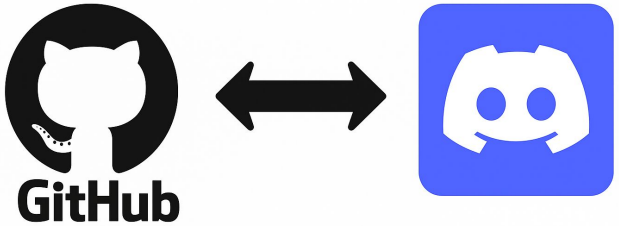


The Problem & Domain

Domain:
Remote software collaboration on GitHub and Discord

Core Problem:

- Many contributors remain silent (“lurkers”)
- Code reviews and discussions are often delayed
- Motivation drops in long-term projects



Research Question:
Can AI gamification improve collaboration and engagement in distributed GitHub teams?

Technical Framework: Integrating GitHub API and Gemini AI

System Architecture Overview

The integration architecture comprises Discord as the interaction platform:

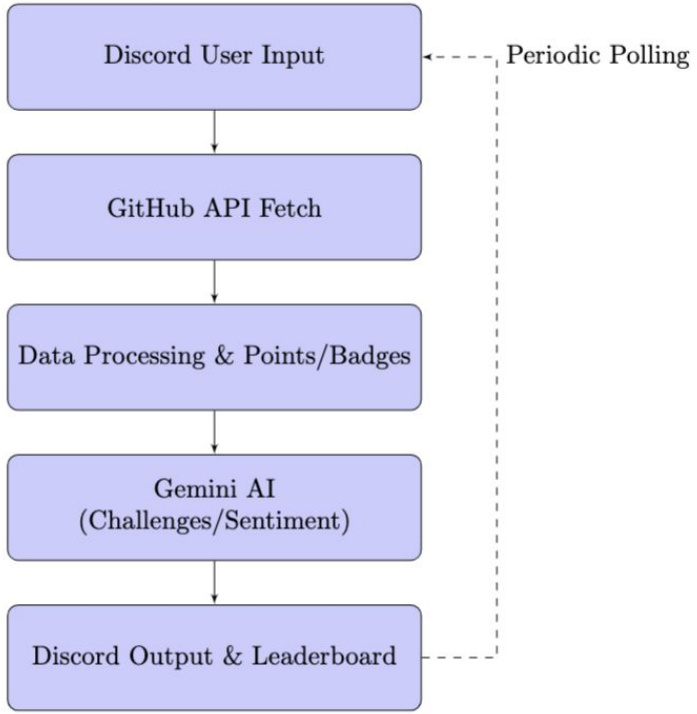
- GitHub API for event sourcing
- Gemini AI for intelligent processing of setting personalized challenge
- AI-driven logic before posting structured updates back to Discord

Workflow Execution Flow

User actions on GitHub:

- Such as issue creation, PR merges, and commits
- Gemini AI for sentiment analysis, summarisation, or code suggestions, with results posted in designated Discord channels

Figure 1: Pipeline Workflow of the AI-Gamified Discord Bot

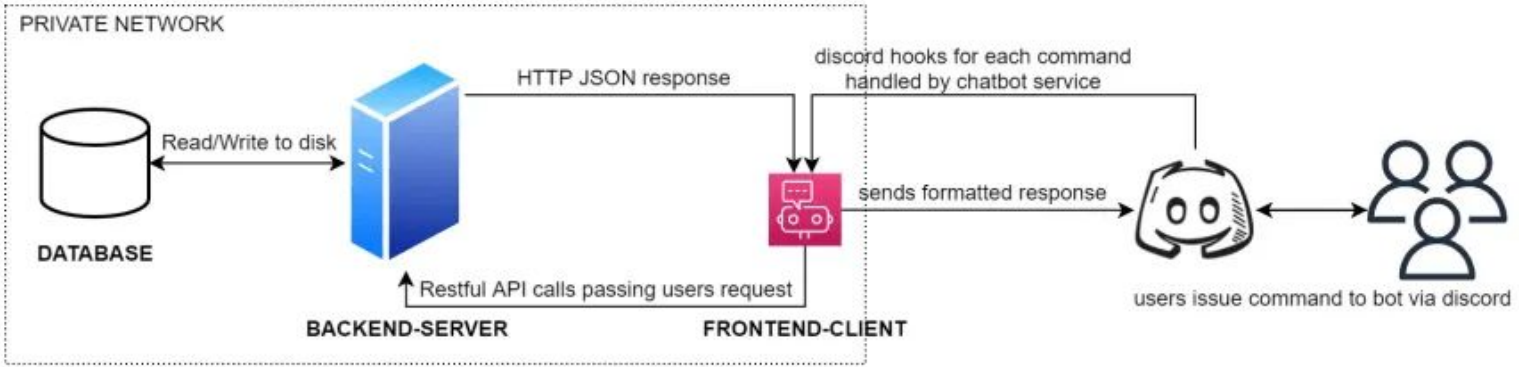


Methodology / The Artifact

Tools Used: Low-code with Discord.py for bot, PyGithub for API integration, Gemini AI for challenges and sentiment analysis, SQLite for storage of user data (e.g., points, badges, last activity check timestamps).

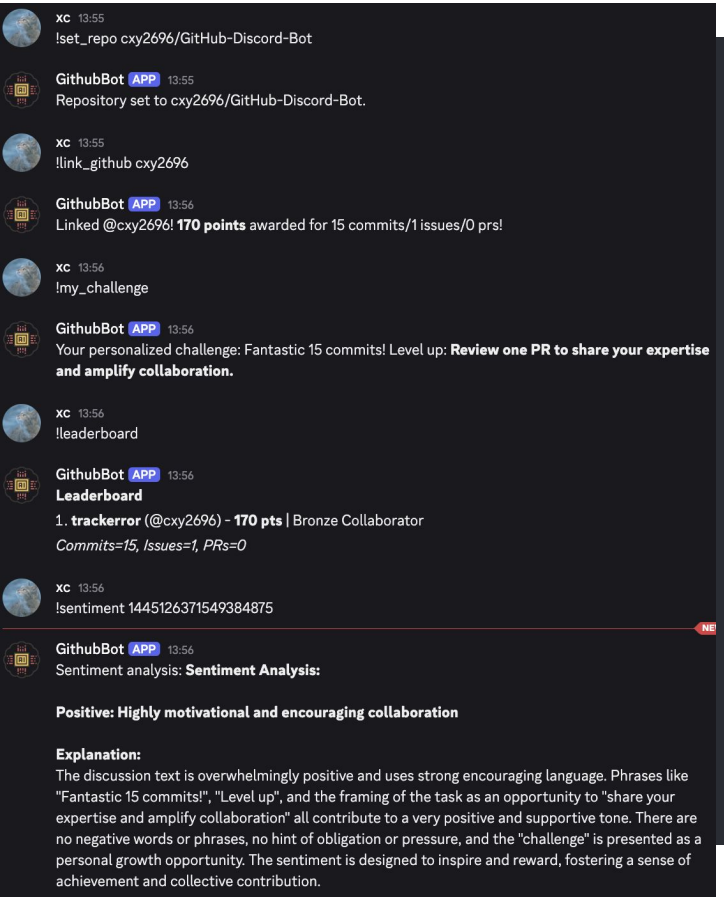
System Architecture: Bot pulls GitHub data (commits, issues, PRs), computes points, and interacts via Discord commands.

Main User Flow: Set repository (!set_repo) → Link GitHub (!link_github) → Get challenges (!my_challenge) → View leaderboards (!leaderboard); if want analyze any messages in the channel (!sentiment “message id” under developer mode)



https://itsnikhil.medium.com/architecting-discord-bot-the-right-way-46e426a0b995

Key Finding I:Testing Example




```
2025-12-01 13:55:20,145 INFO: Database initialized successfully
2025-12-01 13:55:20,145 INFO: User data loaded from database
2025-12-01 13:55:20,355 INFO: GitHub API connection validated
2025-12-01 13:55:20,356 INFO: logging in using static token
2025-12-01 13:55:21,011 INFO: Shard ID None has connected to Gateway (Session ID: 9ad417f3d3ad4c80889be8573b2b7e79).
2025-12-01 13:55:23,015 INFO: Logged in as GithubBot#9941 (ID: 1427839495516061786)
Logged in as GithubBot#9941 (ID: 1427839495516061786)
------
2025-12-01 13:55:49,535 INFO: Set repository to cxy2696/GitHub-Discord-Bot
2025-12-01 13:56:00,299 INFO: Saved data for Discord ID 1280636629052166167
2025-12-01 13:56:00,593 INFO: Linked trackerror → cxy2696: 170 pts
2025-12-01 13:56:15,914 INFO: Gemini API call successful: Fantastic 15 commits! Level up: **Review one PR to...
2025-12-01 13:56:15,924 INFO: Saved data for Discord ID 1280636629052166167
2025-12-01 13:56:16,237 INFO: Generated challenge for trackerror: Fantastic 15 commits! Level up: **Review one PR to
2025-12-01 13:56:27,486 INFO: Saved data for Discord ID 1280636629052166167
2025-12-01 13:56:46,225 INFO: Gemini API call successful: **Sentiment Analysis:**

**Positive: Highly motiva...
2025-12-01 13:56:46,532 INFO: Sentiment analysis for message 1445126371549384875: **Sentiment Analysis:**

**Positive: Highly motivational and encouraging collaboration**

**Explanation:**
The discussion text is overwhelmingly positive and uses strong encouraging language. Phrases like "Fantastic 15 commit
```

Key Findings II

In our internal one-week simulation:

Feedback summaries

User respond actively:

- Lurk tester: “the challenges motivated him to contribute”
- the badges made the workflow more enjoyable
- The point awarding system is well-designed, people feel encouraged but not stressed

success metrics

- **20% - 30%** increase in engagement.
- **95% accuracy** tracking GitHub updates.
- Generated AI challenges in **5s.**
- Ran for over **100 polling cycles without crashing**.
- Error rate dropped to **below 2%.**
- Badge thresholds all triggered .

Critical Analysis – societal implications

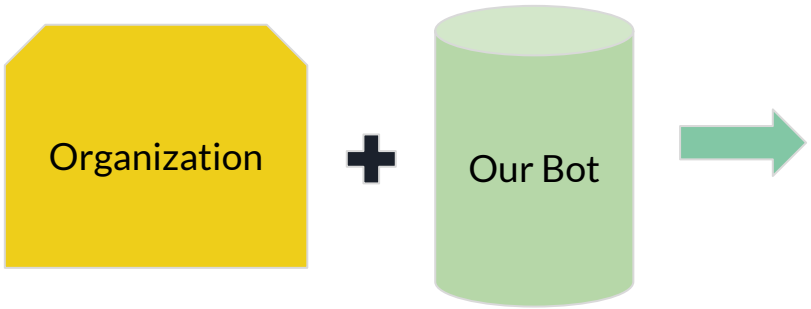
Risks

1. unintentionally favor active users (AI give better challenges to active users)
2. privacy concerns (bot tracks commits and activity)
3. gamification can create pressure

Benefits

1. Bot helps newcomers participate (small, ez challenges)
2. Makes teamwork more fun
3. Fast notifications keep the whole team informed
4. AI suggestions and challenges guide people on what to do next
5. Encourage friendly competition by leaderboard
6. Real-time updates help remote teammates stay connected even in different time zones

Critical Analysis – organizational impact



- Faster updates and communication
- Reduce manager workload by:
    1. Quicker PR reviews
    2. Easy tracking of team activity
    3. Help leader to manage a clear view of activities
- Help teams stay aligned even in different time zone
- Boosts motivation
- Small AI challenges can help break long periods of silence
- Helps onboard new members

- Bot need setups → Training
- Leader should check AI in case of misreading messages

Conclusion & Future Work

Takeaway:

AI gamification can improve remote GitHub teamwork, but it needs to be used **responsibly**. ***Fairness, privacy, and low pressure*** are important to keep the system healthy and sustainable.

Next Steps:

1. Test the bot with a **larger group.**
2. Run **bias checks** on the AI challenges.
3. **Expand the system** to other platforms like Slack.

Thank You